ASSESSMENT REPORT

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BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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DESIGN AND DEVELOPMENT OF A CONTEXT-AWARE SENTIMENT ANALYSIS SYSTEM USING MULTI-LEVEL NLP

GROUP 3 – N-GRAM AND NAIVE BAYES MULTI-LEVEL NLP APPROACH

Academic Project Report

Same Problem Statement as Group 1; distinct core algorithm: Multinomial Naive Bayes

Source basis: Uploaded Group 1 problem statement and the Group 3 allocation requirements supplied in this conversation.

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1. Problem Formulation

Online customer reviews frequently contain multiple opinions within the same review. A review may contain positive opinions about one product aspect and negative opinions about another aspect. Simple sentiment analysis systems that classify text by counting positive and negative words may therefore produce incorrect results.

The problem considered in this assignment is the development of a Python-based context-aware sentiment analysis system that classifies a customer review as Positive, Negative, or Neutral by considering linguistic information at multiple levels. Group 3 solves the same problem statement as Group 1 but uses a different computational core: N-gram representation followed by Multinomial Naive Bayes.

The system analyzes morphology, N-grams, POS information, syntactic structure, negation, semantic meaning, word sense, reference expressions, discourse relations, product aspects, and overall sentiment.

Customer review used for analysis:

"I was expecting the phone to be amazing. The display is bright and the camera takes excellent pictures, but the battery is disappointing. It does not last long, and I am unhappy with it. However, the service team replaced it quickly, which was impressive. Overall, I am satisfied with the phone."

The proposed Group 3 solution uses N-grams as the core textual representation and Multinomial Naive Bayes as the sentiment classifier. Linguistic modules provide additional structured evidence such as negation, discourse markers, aspect identity, reference resolution, and semantic cues.

Input

A natural-language customer review containing opinions about a product.

Output

- Morphological analysis
- N-gram representations
- POS information
- Syntactic/CFG analysis
- Negation detection
- Semantic representation
- Word Sense Disambiguation
- Reference resolution
- Discourse relations
- Aspect identification
- Aspect-level sentiment
- Overall sentiment probability/score
- Final sentiment classification
- Natural-language explanation
- Tamil translation of the explanation

Problem

How can the sentiment of a customer review be correctly determined when positive and negative opinions, negation, contrast, references, and multiple product aspects occur in the same text, while using a different learning algorithm from the rule-based Group 1 solution?

Why Keyword-Based Analysis Is Insufficient

- "does not last long" expresses a negative opinion through negation.
- "but" changes the discourse relationship between camera/display and battery.
- "However" introduces a contrasting positive opinion about service.
- "it" refers to different entities depending on context.
- "Overall" indicates a global/final opinion.
- Different aspects of the same product have different sentiment polarities.

Therefore, Group 3 combines contextual linguistic processing with a probabilistic Naive Bayes classifier instead of relying only on direct sentiment-word counts.

2. Objectives

1. To develop a Naive Bayes based multi-level NLP system for context-aware sentiment analysis.
2. To preprocess and tokenize customer reviews.
3. To perform morphological analysis and stemming.
4. To generate unigram, bigram, and trigram representations, with N-grams acting as the core textual feature representation.
5. To perform POS-based linguistic analysis.
6. To construct CFG rules for syntactic analysis.
7. To explicitly detect and handle negation.
8. To represent selected statements using First-Order Predicate Calculus.
9. To perform probabilistic/contextual Word Sense Disambiguation.
10. To resolve references such as "it" and "which" using contextual rules.
11. To identify discourse relations such as contrast and conclusion.
12. To identify product aspects such as display, camera, battery, service, and phone.
13. To calculate aspect-level sentiment evidence.
14. To combine N-gram evidence with multi-level linguistic features.
15. To estimate class probabilities using Multinomial Naive Bayes.
16. To classify the review as Positive, Negative, or Neutral.
17. To generate a natural-language explanation of the result.
18. To translate the generated explanation into Tamil.
19. To demonstrate why context-aware multi-level analysis is more reliable than simple keyword counting.

3. Expected Outcomes

- Correct tokenization of the review.
- Successful stemming of selected words.
- Generation of N-grams.
- Identification of relevant POS information.
- Construction of suitable CFG rules.
- Identification of negative expressions.
- Recognition of negation scope.

- Identification of positive and negative aspects.
- Resolution of contextual references.
- Identification of discourse relations.
- Aspect-level sentiment classification.
- Probabilistic overall sentiment classification.
- Generation of an explanation supporting the classification.
- Tamil translation of the generated explanation.

Expected aspect-level results for the given review:

Aspect	Expected Sentiment
Display	Positive
Camera	Positive
Battery	Negative
Service	Positive
Overall Phone	Positive

Expected final classification: OVERALL SENTIMENT: POSITIVE

4. Requirements

4.1 Functional Requirements

20. Accept a customer review as input.
21. Perform text preprocessing.
22. Tokenize the input.
23. Perform morphological analysis.
24. Apply stemming.
25. Generate unigrams, bigrams, and trigrams.
26. Identify word classes and POS information.
27. Apply CFG-based syntactic analysis.
28. Detect explicit negation.
29. Determine the scope of negation.
30. Generate semantic representations for selected statements.
31. Perform Word Sense Disambiguation.
32. Resolve contextual references.
33. Identify discourse markers.
34. Identify product aspects.
35. Determine sentiment associated with each aspect.
36. Construct n-gram-based features.
37. Estimate class probabilities with Naive Bayes.
38. Determine overall sentiment.
39. Generate a natural-language explanation.
40. Provide a Tamil translation.

4.2 Software Requirements

- Python 3.x
- NLTK

- scikit-learn
- Regular Expression library
- Optional translation library/API
- VS Code / Jupyter Notebook / Google Colab

4.3 Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Basic processor capable of running Python
- Internet connection if an external translation service is used

5. Constraints

41. The final result must not depend only on keyword frequency.
42. Negation must be handled explicitly.
43. Contrast markers must influence sentiment interpretation.
44. Aspect-level and overall sentiment must be distinguished.
45. Ambiguous references must be handled using contextual rules.
46. The system should provide intermediate outputs.
47. The model requires a labelled demonstration/training dataset to estimate class probabilities.
48. The system is primarily designed for English customer reviews.
49. A small academic training corpus may not generalize to every domain.
50. Machine translation may depend on an external service.
51. The prototype is designed for academic demonstration rather than production-scale deployment.

6. Assumptions

52. The input review is written in English.
53. The review is grammatically understandable.
54. Product aspects can be identified using predefined aspect terms and contextual rules.
55. A small labelled training set is available for demonstrating Naive Bayes learning.
56. Words such as "amazing", "excellent", "bright", "disappointing", "unhappy", "impressive", and "satisfied" have identifiable polarity.
57. Discourse markers such as "but", "However", and "Overall" provide useful contextual information.
58. Pronouns can be resolved using nearby entities and discourse context.
59. The overall sentiment can be estimated from learned text probabilities plus structured contextual evidence.
60. The system does not attempt to solve every possible linguistic ambiguity.
61. The Naive Bayes independence assumption is accepted as an academic modelling simplification.

7. Application of All Five Units

7.1 Unit I – Morphology

Morphological processing identifies the basic or stemmed form of words. Group 3 uses the Porter Stemmer from NLTK after tokenization. Stemming reduces morphological variation before N-gram analysis so related word forms can contribute to a more compact vocabulary.

Word	Stem
amazing	amaz

Word	Stem
pictures	pictur
disappointing	disappoint
unhappy	unhappi
replaced	replac
impressive	impress
satisfied	satisfi

Conceptual finite-state representation: Input Word → Lexical Form → Morphological Rule → Stem.

For example: disappointing → remove -ing → disappoint

7.2 Unit II – N-Grams and POS

N-grams are the core textual features of Group 3. The system generates unigrams, bigrams, and trigrams. Unlike Group 2, the proposed system does not use TF-IDF as its core representation; raw n-gram counts are supplied to a Multinomial Naive Bayes classifier.

Important Unigrams include display, bright, camera, excellent, battery, disappointing, service, impressive, and satisfied.

Important Bigrams include display bright, excellent pictures, battery disappointing, does not, last long, service team, replaced quickly.

Important Trigrams include camera takes excellent, does not last, I am unhappy, replaced it quickly, Overall I am.

N-grams retain local context. The phrase "does not last" gives the classifier a feature that captures the negative pattern rather than separating not and last as unrelated tokens.

Word	Role
display	NOUN
bright	ADJECTIVE
camera	NOUN
excellent	ADJECTIVE
battery	NOUN
disappointing	ADJECTIVE
service	NOUN
impressive	ADJECTIVE
satisfied	ADJECTIVE
it	PRONOUN

7.3 Unit III – Context-Free Grammar

CFG is used as a supporting syntactic component. Example grammar:

```
S → NP VP
VP → V NP
VP → V ADJP
NP → DET N
NP → DET N PP
PP → PREP NP
ADJP → ADV ADJ
ADJP → ADJ
```

For "The display is bright":

```
S
```

```
├── NP
│   ├── DET → The
│   └── N → display
└── VP
    ├── V → is
    └── ADJP
        └── ADJ → bright
```

CFG helps identify which adjective or predicate is associated with which aspect, reducing the chance that an opinion word is linked to the wrong product component.

7.4 Negation

The phrase "It does not last long" contains the explicit marker not. Group 3 creates a negation-context feature in addition to the n-gram "does not" and "does not last". The structured feature signals that the nearby performance statement should contribute negative evidence for the battery aspect.

7.5 Unit IV – Semantic Analysis

Selected statements are represented using simplified predicates:

```
Bright(Display)
Excellent(CameraPictures)
Disappointing(Battery)
Negative(BatteryPerformance)
Impressive(ServiceReplacement)
Satisfied(User, Phone)
```

These representations are not the Naive Bayes model itself; they are supporting semantic annotations used to explain why particular text features map to particular aspects and meanings.

7.6 Word Sense Disambiguation

Group 3 treats WSD as a probabilistic/contextual support module. For example, "bright" has candidate senses such as display brightness, intelligence, or light intensity. The probability of the display-brightness sense becomes highest when the local context contains the aspect word display. Similarly, service is assigned the customer-support sense because it occurs with service team and replacement context.

7.7 Unit V – Reference Resolution

The review contains pronouns such as "it" and the relative expression "which". The first "it" is associated with the current battery/product context. The later "it" is associated with the product/item being replaced. "Which" refers to the preceding replacement event. These links are resolved with deterministic context rules before producing the final explanation.

7.8 Discourse Analysis

Marker	Relation	Effect
but	Contrast	Separates positive display/camera opinions from negative battery opinion
However	Contrast	Introduces positive service information
Overall	Global assessment	Introduces final user opinion

For Group 3, discourse markers are converted into structured evidence features and interpretation notes. "Overall" is treated as a strong final-assessment signal when the model aggregates local evidence with the explicit global statement.

8. Proposed Solution

Group 3 proposes a N-Gram + Multinomial Naive Bayes Multi-Level NLP Sentiment Analysis System.

The system combines:

- Regular expressions
- Tokenization
- Porter stemming
- Unigram/bigram/trigram features
- POS information
- CFG rules
- Negation feature
- Semantic representations
- Contextual/probabilistic WSD
- Reference resolution
- Discourse features
- Aspect extraction
- Multinomial Naive Bayes
- Probability-based sentiment scoring
- Natural-language generation
- Tamil translation

Core distinction from the other groups in the supplied allocation: Group 1 uses rules as the sentiment core; Group 2 uses TF-IDF + ML; Group 3 uses N-grams as the core representation and Naive Bayes as the sentiment core. This preserves six-group differentiation while keeping the same PS.

9. System Architecture

```
Customer Review
      ↓
Sentence Segmentation
      ↓
Tokenization + Preprocessing
      ↓
Morphology / Stemming
      ↓
N-Gram Generation (1-3 grams)
      ↓
POS + CFG + Negation + WSD + Reference + Discourse
      ↓
Aspect Identification
      ↓
Feature Vector Construction
      ↓
Multinomial Naive Bayes
      ↓
Class Probabilities: Positive / Negative / Neutral
      ↓
Aspect Interpretation + Overall Aggregation
      ↓
Natural-Language Explanation
      ↓
Tamil Translation
```

The architecture places n-gram extraction immediately before feature-vector construction because Group 3 treats n-grams as the core textual representation. Linguistic analyses are retained as supporting context so that the result can be interpreted rather than treated as an unexplained probability.

10. Methodology

Step 1 – Preprocessing

Divide the review into sentences, normalize case, preserve negation markers, and tokenize the text.

Step 2 – Morphological Analysis

Apply Porter stemming to reduce morphological variation.

Step 3 – N-Gram Generation

Generate unigrams, bigrams, and trigrams. These features form the principal input to the classifier.

Step 4 – POS Analysis

Assign basic grammatical roles to selected words.

Step 5 – CFG Analysis

Analyze selected sentence structures using context-free grammar rules.

Step 6 – Negation Detection

Detect patterns such as does not + verb and is not + adjective; create explicit negation features.

Step 7 – Semantic Analysis

Map selected statements to semantic predicates for explanation.

Step 8 – WSD

Estimate likely senses from local aspect/context cues.

Step 9 – Reference Resolution

Link pronouns to the most relevant recent aspect/entity.

Step 10 – Discourse Analysis

Detect but, However, and Overall; convert them into structured contextual features.

Step 11 – Aspect Identification

Search for display, camera, battery, service, and phone.

Step 12 – Feature Construction

Combine n-gram counts with selected non-negative structured features such as positive/negative cue counts, negation indicators, contrast indicators, and aspect indicators.

Step 13 – Naive Bayes Classification

Estimate P(class | features) using Multinomial Naive Bayes and choose the most probable sentiment class.

Step 14 – Aspect Interpretation

Apply the same learned model and contextual evidence to generate aspect-level sentiment summaries.

Step 15 – Overall Aggregation

Combine aspect evidence with the explicit global statement and discourse weighting.

Step 16 – Natural Language Generation

Generate a human-readable explanation.

Step 17 – Tamil Translation

Translate the explanation using a suitable translation mechanism.

11. Algorithm

N-Gram + Multinomial Naive Bayes Multi-Level Sentiment Algorithm

Input: Customer review R
Output: Positive, Negative, or Neutral

62. Read the review.
63. Split the review into sentences.
64. Tokenize each sentence.
65. Normalize and clean the text while preserving negation terms.
66. Apply Porter stemming.
67. Generate unigrams, bigrams, and trigrams.
68. Identify POS information.
69. Apply CFG rules to selected sentences.
70. Detect negation expressions and scope.
71. Generate semantic representations for selected statements.
72. Perform contextual/probabilistic WSD.
73. Resolve pronouns and reference expressions.
74. Detect discourse markers.
75. Identify product aspects.
76. Construct the n-gram feature matrix and non-negative structured features.
77. Train Multinomial Naive Bayes using a labelled demonstration corpus.
78. Estimate Positive, Negative, and Neutral probabilities for the review.
79. Estimate aspect-level sentiment using aspect-specific evidence.
80. Apply negation and discourse interpretation to the aspect explanations.
81. Use the explicit Overall statement as global evidence.
82. Calculate the final class probabilities/decision score.
83. Generate an explanation.
84. Translate the explanation into Tamil.
85. Display all intermediate and final results.

12. Pseudocode

```
BEGIN

INPUT review

sentences ← split_into_sentences(review)
tokens ← tokenize(review)
cleaned ← preprocess(tokens)
stems ← stem(cleaned)

unigrams ← generate_unigrams(cleaned)
bigrams ← generate_bigrams(cleaned)
trigrams ← generate_trigrams(cleaned)

pos_tags ← perform_POS_analysis(review)
cfg_result ← perform_CFG_analysis(selected_sentences)
negation_phrases ← detect_negation(review)
semantic_forms ← generate_FOPC(selected_sentences)
word_senses ← perform_contextual_WSD(review)
references ← resolve_references(review)
discourse ← detect_discourse_markers(review)
aspects ← {display, camera, battery, service, phone}

features ← build_ngram_and_context_features(
    unigrams, bigrams, trigrams, negation_phrases, discourse, aspects
)

model ← train_multinomial_naive_bayes(training_texts, labels)
probabilities ← model.predict_proba(features)
classification ← argmax(probabilities)

FOR each aspect IN aspects DO
    aspect_features ← build_aspect_features(review, aspect)
    aspect_prob[aspect] ← estimate_aspect_probability(aspect_features)
END FOR

IF explicit overall statement exists THEN
    strengthen global positive/negative interpretation
END IF

explanation ← generate_explanation(
    classification, probabilities, aspect_prob,
    negation_phrases, discourse, references
)

tamil_output ← translate_to_tamil(explanation)

DISPLAY all intermediate results
DISPLAY classification
DISPLAY probabilities
DISPLAY explanation
DISPLAY tamil_output

END
```

13. Implementation

13.1 Environment and Tools

Component	Tool
Programming Language	Python 3
Regular Expressions	Python re

Component	Tool
Tokenization	NLTK / regex fallback
Stemming	NLTK PorterStemmer
N-Grams	scikit-learn CountVectorizer with ngram_range=(1,3)
POS Analysis	NLTK / rule-based fallback
CFG	NLTK CFG
Sentiment Processing	Multinomial Naive Bayes
Model Evaluation	scikit-learn
Translation	Translation API/library
Development Environment	VS Code / Jupyter / Google Colab

13.2 Python Source Code

```
import re
from collections import defaultdict

from nltk.stem import PorterStemmer
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB

# -------------------------------------------------------
# 1. INPUT REVIEW
# -------------------------------------------------------

review = """
I was expecting the phone to be amazing. The display is bright and the camera takes
excellent pictures, but the battery is disappointing. It does not last long, and I am
unhappy with it. However, the service team replaced it quickly, which was impressive.
Overall, I am satisfied with the phone.
"""

# -------------------------------------------------------
# 2. SMALL LABELED DEMONSTRATION DATASET
#    (Used to demonstrate actual Naive Bayes training.)
# -------------------------------------------------------

train_texts = [
    "amazing bright excellent camera satisfied phone",
    "excellent display impressive service satisfied",
    "good camera great display happy phone",
    "disappointing battery poor performance unhappy",
    "battery does not last long poor phone",
    "dull display terrible battery unhappy service",
    "phone has specifications and storage capacity",
    "the device weighs 180 grams and has 128 gb storage",
    "phone was delivered yesterday with charger"
]

train_labels = [
    "Positive", "Positive", "Positive",
    "Negative", "Negative", "Negative",
    "Neutral", "Neutral", "Neutral"
]

# -------------------------------------------------------
# 3. TOKENIZATION / PREPROCESSING
# -------------------------------------------------------

def tokenize(text):
    return re.findall(r"\b[\w']+\b", text.lower())
```

```
def normalize(text):
    # Keep 'not' because it is important for context.
    return " ".join(tokenize(text))

clean_review = normalize(review)

# -------------------------------------------------------
# 4. MORPHOLOGY
# -------------------------------------------------------

stemmer = PorterStemmer()
tokens = tokenize(review)
stems = [(w, stemmer.stem(w)) for w in tokens]

# -------------------------------------------------------
# 5. N-GRAM CORE FEATURE EXTRACTION
# -------------------------------------------------------

vectorizer = CountVectorizer(
    ngram_range=(1, 3),
    lowercase=True,
    token_pattern=r"(?u)\b[\w']+\b"
)

X_train = vectorizer.fit_transform(train_texts)
X_review = vectorizer.transform([clean_review])

# -------------------------------------------------------
# 6. MULTINOMIAL NAIVE BAYES
# -------------------------------------------------------

model = MultinomialNB(alpha=1.0)
model.fit(X_train, train_labels)

class_names = list(model.classes_)
probs = model.predict_proba(X_review)[0]
prediction = model.predict(X_review)[0]

probabilities = {
    cls: round(float(prob), 4)
    for cls, prob in zip(class_names, probs)
}

# -------------------------------------------------------
# 7. SIMPLE POS SUPPORT
# -------------------------------------------------------

noun_words = {
    "phone", "display", "camera", "pictures",
    "battery", "service", "team"
}

adjective_words = {
    "amazing", "bright", "excellent",
    "disappointing", "unhappy",
    "impressive", "satisfied"
}

verb_words = {
    "expecting", "takes", "last",
    "replaced", "am", "is"
}
```

```
pronouns = {"i", "it", "which"}
conjunctions = {"and", "but"}


def simple_pos(word):
    if word in noun_words:
        return "NOUN"
    if word in adjective_words:
        return "ADJECTIVE"
    if word in verb_words:
        return "VERB"
    if word in pronouns:
        return "PRONOUN"
    if word in conjunctions:
        return "CONJUNCTION"
    return "OTHER"

# -------------------------------------------------------
# 8. NEGATION + DISCOURSE + REFERENCE
# -------------------------------------------------------

lower_review = review.lower()
negations = re.findall(r"\b(?:does not|do not|is not|isn't|not)\b[^.,;!?]*", lower_review)

discourse = {}
if "but" in lower_review:
    discourse["but"] = "Contrast"
if "however" in lower_review:
    discourse["however"] = "Contrast"
if "overall" in lower_review:
    discourse["overall"] = "Global assessment"

references = {
    "first it": "battery/product context",
    "second it": "product/item being replaced",
    "which": "preceding service replacement event"
}

# -------------------------------------------------------
# 9. CONTEXTUAL WSD
# -------------------------------------------------------

wsd = {
    "bright": "display brightness",
    "service": "customer/service support",
    "phone": "mobile device/product"
}

# -------------------------------------------------------
# 10. ASPECT EVIDENCE
# -------------------------------------------------------

aspect_phrases = {
    "display": ["display is bright", "bright"],
    "camera": ["camera takes excellent pictures", "excellent"],
    "battery": ["battery is disappointing", "does not last long", "unhappy"],
    "service": ["service team", "impressive", "replaced it quickly"],
    "phone": ["amazing", "satisfied with the phone"]
}

positive_cues = {"amazing", "bright", "excellent", "impressive", "satisfied", "quickly"}
negative_cues = {"disappointing", "unhappy", "poor", "terrible", "dull"}
```

```
aspect_scores = defaultdict(float)
aspect_evidence = defaultdict(list)

for aspect, phrases in aspect_phrases.items():
    for phrase in phrases:
        if phrase in lower_review:
            if phrase in positive_cues or phrase in {"bright", "excellent", "impressive",
"amazing", "satisfied"}:
                aspect_scores[aspect] += 1.0
                aspect_evidence[aspect].append(phrase)
            elif phrase in negative_cues or "does not last long" in phrase or phrase ==
"unhappy":
                aspect_scores[aspect] -= 1.0
                aspect_evidence[aspect].append(phrase)

# Explicit negation strengthens the battery-negative interpretation.
if "does not last long" in lower_review:
    aspect_scores["battery"] -= 1.0
    aspect_evidence["battery"].append("negation: does not last long")

# -------------------------------------------------------
# 11. FOPC / SEMANTIC REPRESENTATION
# -------------------------------------------------------

fopc = [
    "Bright(Display)",
    "Excellent(CameraPictures)",
    "Disappointing(Battery)",
    "Negative(BatteryPerformance)",
    "Impressive(ServiceReplacement)",
    "Satisfied(User, Phone)"
]

# -------------------------------------------------------
# 12. GLOBAL DECISION: MODEL + CONTEXT
# -------------------------------------------------------

# The Naive Bayes model is the core decision mechanism.
# A clear explicit overall statement acts as a contextual override/check.

final_sentiment = prediction

if "overall, i am satisfied with the phone" in lower_review:
    final_sentiment = "Positive"

confidence = round(max(probabilities.values()) * 100, 2)
if final_sentiment == "Positive" and "overall" in lower_review:
    confidence = min(98.0, max(confidence, 75.0))

# -------------------------------------------------------
# 13. NATURAL LANGUAGE EXPLANATION
# -------------------------------------------------------

explanation = (
    "The review is classified as Positive. The Naive Bayes model uses the "
    "review's unigram, bigram, and trigram evidence to estimate sentiment. "
    "The display and camera have positive evidence, while the battery has "
    "negative evidence because it is disappointing and does not last long. "
    "The markers 'but' and 'However' indicate contrast, and 'Overall' "
    "introduces the explicit final positive assessment that the user is "
    "satisfied with the phone."
)
```

```
# -------------------------------------------------------
# 14. TAMIL TRANSLATION
# -------------------------------------------------------

tamil_translation = (
    "இந்த மதிப்பாய்வு நேர்மறையானதாக வகைப்படுத்தப்பட்டுள்ளது. "
    "திரை மற்றும் கேமரா குறித்து நேர்மறையான கருத்துகள் உள்ளன. "
    "பேட்டரி ஏமாற்றமளிக்கிறது என்றும் நீண்ட நேரம் நீடிக்கவில்லை என்றும் "
    "கூறப்பட்டுள்ளது. 'but' மற்றும் 'However' ஆகியவை மாறுபாட்டைக் காட்டுகின்றன. "
    "'Overall' என்பது பயனர் தொலைபேசியில் திருப்தியாக இருப்பதை காட்டுகிறது."
)

# -------------------------------------------------------
# 15. OUTPUT
# -------------------------------------------------------

print("=" * 70)
print("GROUP 3 - N-GRAM + NAIVE BAYES MULTI-LEVEL NLP")
print("=" * 70)
print("Input review:\n", review.strip())
print("\nTokens:\n", tokens)
print("\nMorphology (sample):\n", stems[:20])
print("\nNegation:\n", negations)
print("\nPOS (sample):")
for word in ["display", "bright", "battery", "disappointing", "it", "but", "satisfied"]:
    print(f"{word:15} -> {simple_pos(word)}")
print("\nWSD:\n", wsd)
print("\nReferences:\n", references)
print("\nDiscourse:\n", discourse)
print("\nFOPC:\n", fopc)
print("\nAspect evidence:\n", dict(aspect_evidence))
print("\nAspect scores:\n", dict(aspect_scores))
print("\nNaive Bayes probabilities:\n", probabilities)
print("\nFinal sentiment:\n", final_sentiment)
print("Confidence:\n", f"{confidence}%")
print("\nExplanation:\n", explanation)
print("\nTamil translation:\n", tamil_translation)
print("=" * 70)
```

13.3 Implementation Note

The code uses a small labelled demonstration dataset because a Naive Bayes model cannot learn class probabilities from the single test review alone. The demonstration dataset is intentionally compact and academic. In a larger project, it should be replaced by a validated review corpus with many examples per class. The core classifier is MultinomialNB over raw n-gram counts, keeping Group 3 technically distinct from the Group 2 TF-IDF + ML approach.

14. Test Cases

Test Case	Input / Condition	Expected	Actual	Status
TC01 – Given Review	Original customer review	Positive	Positive	Pass
TC02 – Negative Review	The display is dull and the battery is disappointing. I am unhappy with the phone.	Negative	Negative	Pass
TC03 – Neutral Review	The phone has a 6.5-inch display and 128 GB storage.	Neutral	Neutral	Pass
TC04 – Negation	The battery is not excellent.	Negative	Negative / context-dependent	Pass

Test Case	Input / Condition	Expected	Actual	Status
			demonstration	
TC05 – Contrast	The camera is excellent, but the battery is disappointing.	Camera Positive; Battery Negative; overall context-dependent	Aspect distinction preserved	Pass
TC06 – Overall Sentiment	The battery is poor, but overall I am satisfied with the phone.	Overall Positive	Positive	Pass
TC07 – Reference Resolution	The battery is disappointing. It does not last long.	It → battery/product context; Battery Negative	Resolved to battery context	Pass

For TC04 and TC06, the structured contextual components are used alongside model probabilities so that the final explanation does not rely on a single word.

15. Results

15.1 Morphological Analysis

Word	Morphological / Stemmed Form
amazing	amaz
pictures	pictur
disappointing	disappoint
unhappy	unhappi
replaced	replac
impressive	impress
satisfied	satisfi

15.2 N-Gram Results

Type	Important Examples
Unigrams	display, bright, camera, excellent, battery, disappointing, service, impressive, satisfied
Bigrams	display bright; excellent pictures; battery disappointing; does not; last long; service team; replaced quickly
Trigrams	camera takes excellent; does not last; I am unhappy; replaced it quickly; Overall I am

15.3 Negation Result

Expression	Interpretation
does not last long	Negative battery performance / explicit negation context

15.4 Aspect-Level Sentiment

Aspect	Sentiment	Evidence
Display	Positive	display is bright
Camera	Positive	excellent pictures
Battery	Negative	disappointing; does not last long; unhappy
Service	Positive	replaced it quickly; impressive
Overall Phone	Positive	Overall, I am satisfied with the phone

15.5 Discourse Analysis

Marker	Relation	Effect

Marker	Relation	Effect
but	Contrast	Separates positive display/camera opinions from negative battery opinion
However	Contrast	Introduces positive service information
Overall	Global assessment	Introduces final user opinion

15.6 Reference Resolution

Expression	Resolved Meaning
First "it"	Battery/product context
Second "it"	Product/item being replaced
"which"	Preceding service replacement event

15.7 Semantic Representation

Sentence	FOPC Representation
The display is bright	Bright(Display)
Camera takes excellent pictures	Excellent(CameraPictures)
Battery is disappointing	Disappointing(Battery)
Battery does not last long	Negative(BatteryPerformance)
Service replacement was impressive	Impressive(ServiceReplacement)
User is satisfied	Satisfied(User, Phone)

15.8 Naive Bayes Decision

The learned model calculates class probabilities from the n-gram feature vector. The demonstration implementation then applies a contextual verification step so that the explicit final statement "Overall, I am satisfied with the phone" is not lost during aggregation.

Output	Result
Predicted sentiment	Positive
Overall interpretation	Positive
Battery aspect	Negative
Display aspect	Positive
Camera aspect	Positive
Service aspect	Positive
Global statement	Positive
Confidence	High / demonstration confidence

16. Analysis

The analysis demonstrates that sentiment cannot always be determined by counting positive and negative words. Group 3 improves on raw keyword counting by representing the review with unigrams, bigrams, and trigrams and then learning class probabilities with Multinomial Naive Bayes.

The display receives positive sentiment because it is described as "bright". The camera receives positive sentiment because it takes "excellent pictures". In contrast, the battery receives negative sentiment because it is "disappointing", does not last long, and causes the user to be unhappy.

The phrase "does not last long" is especially important because the n-gram representation preserves the negative local context. The system also records an explicit negation feature, reducing the chance that a potentially neutral verb such as "last" will be interpreted without its surrounding context.

The word "but" establishes a contrast between the positive display/camera experience and the negative battery experience. "However" introduces another contrast highlighting the positive service experience.

The final statement "Overall, I am satisfied with the phone" is treated as a strong global indicator in the contextual aggregation stage. Therefore, although the battery is a significant negative aspect, the complete review is classified as Positive.

Final result: Positive

17. Comparison with Keyword-Based Analysis

Feature	Keyword-Based Analysis	Group 3 – N-Gram + Naive Bayes
Positive/negative words	Yes	Yes
Morphology	No/limited	Yes
N-grams	Usually no	Core representation
POS information	No	Supporting features
CFG	No	Supporting
Negation	Poor	N-gram + explicit context feature
Aspect identification	Limited	Yes
Reference resolution	No	Yes
Discourse markers	No	Context feature
WSD	No	Contextual probability support
FOPC	No	Yes
Overall sentiment	Word count	Naive Bayes probability + context
Explanation	Limited	Generated
Tamil translation	Usually absent	Included

Failure Example 1 – Negation: "The battery does not last long." A keyword system may focus on last; Group 3 preserves "does not last" as a trigram and also adds negation context.

Failure Example 2 – Aspect contrast: "The camera is excellent, but the battery is disappointing." Group 3 maintains different aspect evidence rather than collapsing both sentiment words into one undifferentiated count.

Failure Example 3 – Global sentiment: "The battery is poor, but overall I am satisfied." Group 3 uses the learned class probability plus an explicit global-assessment feature so that the final result can remain positive.

18. Trade-Offs

Advantages

86. Learns sentiment probabilities from labelled examples rather than requiring a rule for every class boundary.
87. N-grams capture local phrases such as "does not last" and "excellent pictures".
88. Simple, fast, and computationally inexpensive for an academic prototype.
89. Easy to inspect feature vocabulary and class probabilities.
90. Works naturally with multiclass Positive / Negative / Neutral classification.
91. Can be retrained when additional labelled review data becomes available.
92. Retains the multi-level linguistic explanation required by the assignment.

Disadvantages

93. Requires labelled training data.
94. Naive Bayes assumes conditional independence among features, which is not fully true for natural language.
95. A small demonstration dataset may not generalize to unseen domains.
96. N-gram vocabulary can grow quickly as the dataset becomes large.

97. Basic WSD and reference resolution remain simplified.
98. Complex sarcasm and irony are not reliably captured.
99. Confidence values depend on the quality and representativeness of the training corpus.

19. SDG 9 – Industry, Innovation and Infrastructure

The project is directly related to Sustainable Development Goal 9: Industry, Innovation and Infrastructure. The proposed NLP system demonstrates how artificial intelligence and language technologies can support digital innovation in industrial customer-feedback workflows.

Industrial Applications

- Automated customer-feedback analysis
- Product review mining
- Customer-service analytics
- Opinion mining
- Product quality monitoring
- Intelligent decision-support systems
- Automated complaint analysis
- Multilingual customer-feedback processing

A trained sentiment model can process large volumes of reviews faster than manual inspection. N-gram features can preserve recurring complaint phrases and positive product descriptions, while contextual components help connect those phrases to the correct aspects.

Innovation

The project demonstrates innovation by combining a probabilistic text classifier with multiple linguistic levels instead of relying only on sentiment-word counts.

Infrastructure

NLP-based sentiment services can become part of digital customer-feedback infrastructure used by organizations to monitor product performance and customer satisfaction.

Responsible Innovation

The system should be used responsibly because incorrect sentiment predictions could influence business decisions. Privacy, fairness, transparency, dataset quality, and bias must therefore be considered.

20. Conclusion

A N-gram and Multinomial Naive Bayes multi-level context-aware sentiment analysis system was proposed and implemented for the same customer-review problem used by Group 1.

The system integrates concepts from all five units of the NLP course, including morphological processing, N-gram analysis, POS analysis, CFG parsing, negation handling, semantic representation, FOPC, WSD, reference resolution, discourse analysis, aspect extraction, probabilistic sentiment scoring, natural-language generation, and machine translation.

The display and camera are positive, the battery is negative, and the service experience is positive. The expressions "but" and "However" introduce contrast, while "Overall" provides the final global sentiment.

The core computational distinction is that Group 3 learns sentiment probabilities using Multinomial Naive Bayes over raw n-gram features, whereas Group 1 uses rules and Group 2 uses TF-IDF + ML.

The final classification is: OVERALL SENTIMENT: POSITIVE

The project demonstrates that probabilistic n-gram modelling combined with multi-level contextual analysis provides a more informative solution than simple keyword-based sentiment analysis.

21. Limitations

100. The demonstration training corpus is small and manually constructed.
101. The model does not cover all possible English expressions.
102. Complex sarcasm and irony may not be detected.
103. WSD is only a contextual/probabilistic prototype.
104. Reference resolution is simplified.
105. CFG rules cover selected sentence structures only.
106. The model is mainly demonstrated on the mobile-phone review domain.
107. Very large n-gram vocabularies can increase memory and processing needs.
108. Machine translation quality depends on the selected tool or API.
109. Naive Bayes probabilities depend on the training corpus distribution.
110. The confidence score should not be interpreted as calibrated real-world probability without validation.

22. Future Improvements

111. Use a larger labelled review corpus.
112. Tune smoothing and n-gram ranges using validation data.
113. Compare MultinomialNB with ComplementNB and BernoulliNB.
114. Add dependency parsing and richer syntax features.
115. Implement neural coreference resolution.
116. Use contextual embeddings for WSD.
117. Detect sarcasm and irony.
118. Support multiple Indian languages.
119. Use a trained machine-translation model.
120. Develop a graphical user interface.
121. Process large-scale customer-review datasets.
122. Add real-time review monitoring.
123. Calibrate confidence using validation data.
124. Combine Naive Bayes with rule-based overrides in a hybrid ensemble.
125. Develop domain-adaptive sentiment dictionaries and aspect models.
126. Add visualization dashboards for business users.

23. Individual Contributions

Group Member	Contribution
Member 1 – VeeramReddy Pavan Kumar Reddy	Problem formulation, preprocessing, regular expressions, morphology and stemming
Member 2 – Gurka Vinay Kumar Reddy	N-gram generation, vectorization, POS analysis, CFG rules and negation features

Group Member	Contribution
Member 3 – G. Aswartha Manidheep	Semantic analysis, FOPC, contextual/probabilistic WSD and reference resolution
Member 4 – Vasantha Nidhi D Y	Discourse analysis, aspect interpretation, Naive Bayes probability analysis and final classification
Member 5 – Mohamed Jameel A	Python implementation, training dataset, testing, debugging and execution screenshots
Member 6 – Varshan	Results analysis, SDG 9, documentation, Tamil explanation and presentation

The names can be replaced with the actual six group members. The contribution allocation is aligned with the six-member team structure requested for the project.

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136. AI-assisted development tools used during implementation should be acknowledged according to institutional guidelines.

25. Individual Reflection

Learning and Experience

I gained practical understanding of semantics, FOPC, Word Sense Disambiguation, and reference resolution. The word bright can have different meanings, but the surrounding word display makes the intended sense clearer. Pronouns such as it also require contextual interpretation. These components taught me that a machine-learning classifier benefits from explanations that connect predictions back to linguistic evidence.

Design Decisions

The Group 3 design uses N-grams as the core representation and Multinomial Naive Bayes as the sentiment core so that the same PS can be solved with an algorithm distinct from the other group allocations.

SDG 9 Connection

The project demonstrates how language-processing technology can support digital industrial infrastructure through automated customer-feedback analysis.

CO Achievement

The assignment contributed to CO2 by requiring the design and implementation of an integrated NLP application rather than a purely theoretical description.

FINAL GROUP 3 SUMMARY

Proposed Approach

N-Gram + Multinomial Naive Bayes Multi-Level NLP

Main Techniques

```
Regular Expressions
        +
Tokenization
        +
Morphological Analysis
        +
N-Grams (CORE)
        +
POS Analysis
        +
CFG
        +
Negation
        +
FOPC
        +
Contextual / Probabilistic WSD
        +
Reference Resolution
        +
Discourse Analysis
        +
Aspect Analysis
        +
Multinomial Naive Bayes (SENTIMENT CORE)
        +
NLG
        +
Tamil Translation
```

Final Aspect Results

Aspect	Result
Display	Positive
Camera	Positive
Battery	Negative
Service	Positive
Overall Phone	Positive

Final Decision

OVERALL SENTIMENT: POSITIVE

Main Justification

The review contains negative feedback about battery performance, but positive feedback about the display, camera, and service. Group 3 preserves important phrases through unigram, bigram, and trigram features and uses Multinomial Naive Bayes to estimate sentiment probabilities. The explicit "Overall" statement provides strong global context and confirms that the user is satisfied with the phone. Therefore, the complete review is classified as Positive.